

Resultant of Coplanar Non-Concurrent Force Systems

High-Level Overview

In Statics and Engineering Mechanics, a **force system** consists of multiple forces and/or moments acting simultaneously on a body. When these forces act in a single 2D plane, they are called **coplanar**. If their lines of action do not all intersect at a single point, they form a **coplanar non-concurrent force system**.

To analyze the combined effect of such a system, we reduce all forces and moments into a single equivalent force called the **Resultant Force (R)**, acting at a specific line of action (or perpendicular distance d) relative to a given reference point (such as the origin or base O).

Fundamental Terminology

- Force Vector ($\mathbf{F}$):** A quantity having both magnitude and direction, measured in Newtons (N) or Kilonewtons (kN).
- Moment (M):** The rotational effect produced by a force about a specific point, measured in Newton-meters (N · m) or Kilonewton-meters (kN · m).

$$\text{Moment} = \text{Force} \times \text{Perpendicular Distance}$$

- Resultant Force (R):** A single force that produces the exact same translational effect as the combined system of original forces.
- Varignon's Theorem (Principle of Moments):** The algebraic sum of moments produced by individual forces about any point is equal to the moment produced by their single resultant force about that same point.

Step-by-Step Analytical Breakdown

To determine the resultant force and its point/distance of application for a coplanar force system, follow this standardized procedure:

1. Vector Resolution of Forces

Decompose every inclined force acting at an angle θ into horizontal (x) and vertical (y) components:

- Horizontal component: $F_x = F \cos \theta$
- Vertical component: $F_y = F \sin \theta$

Sign Conventions:

- Rightward forces ($\rightarrow$) are positive (+); leftward forces ($\leftarrow$) are negative (−).
- Upward forces ($\uparrow$) are positive (+); downward forces ($\downarrow$) are negative (−).

2. Summation of Force Components

Calculate the algebraic sum of all horizontal and vertical forces:

$$\Sigma F_x = \text{Sum of all horizontal force components}$$

$$\Sigma F_y = \text{Sum of all vertical force components}$$

3. Magnitude and Angle of Resultant

The total magnitude R of the resultant force is found using the Pythagorean theorem:

$$R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2}$$

The angle θ that the resultant makes with the horizontal axis is given by:

$$\theta = \tan^{-1} \left(\left| \frac{\Sigma F_y}{\Sigma F_x} \right| \right)$$

4. Moment Summation & Application of Varignon's Theorem

Choose a reference point (e.g., origin O or point A). Compute the net moment ΣM_O^F exerted by all individual forces and explicit moment couples about that point:

Sign Conventions for Moments:

- Counter-Clockwise (CCW) moment = Positive (+)
- Clockwise (CW) moment = Negative (−)

Apply Varignon's Theorem to find the perpendicular distance d from the reference point to the line of action of R :

$$\Sigma M_O^F = R \times d \implies d = \frac{|\Sigma M_O^F|}{R}$$

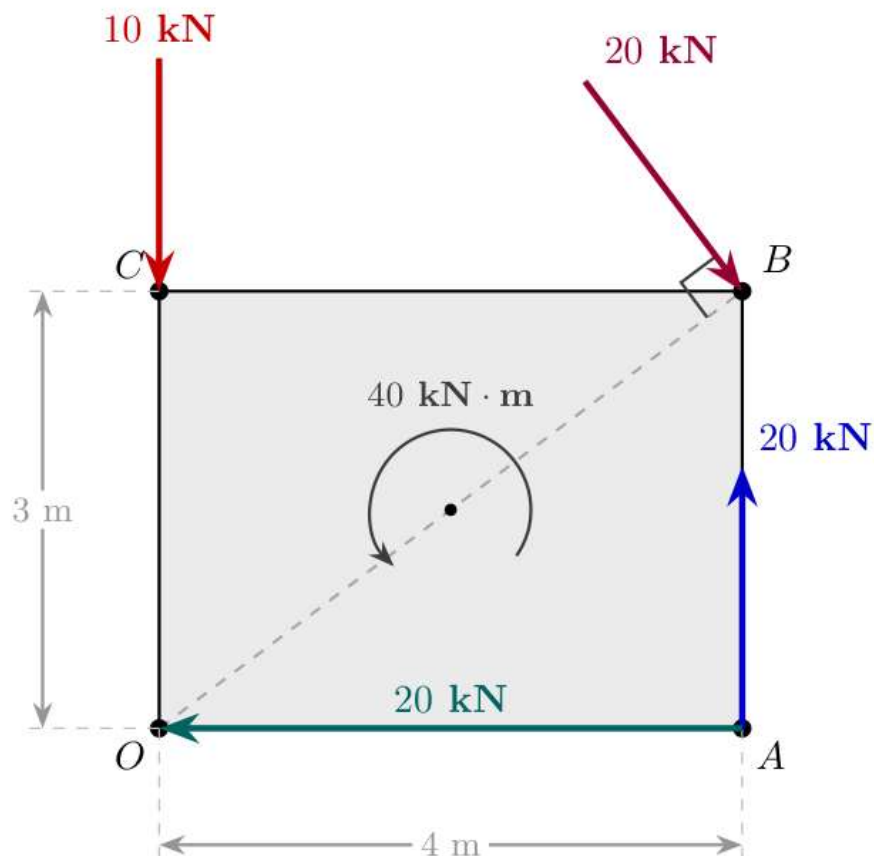
Worked Examples

Problem 1: Rectangular Body $OABC$ Under Multiple Forces and a Couple Moment

Problem Statement

Find the resultant of the force system acting on a body $OABC$ shown in the schematic. Determine the perpendicular distance of the resultant from point O .

- Dimensions: Width $OA = BC = 4 \text{ m}$, Height $OC = AB = 3 \text{ m}$.
- Point O is located at $(0, 0)$.
- Point A is located at $(4, 0)$.
- Point B is located at $(4, 3)$.
- Point C is located at $(0, 3)$.
- Applied Forces/Moments:
 - i. A downward force of 10 kN at point C .
 - ii. An upward force of 20 kN at point A .
 - iii. A leftward force of 20 kN along segment OA at point O .
 - iv. An inclined force of 20 kN acting inward at vertex B , perpendicular to diagonal OB .
 - v. A counter-clockwise couple moment of $40 \text{ kN} \cdot \text{m}$ at the center.



Visual Description: A rectangular body $OABC$ in the first quadrant of a Cartesian plane. Corner O is at $(0, 0)$, A at $(4, 0)$, B at $(4, 3)$, and C at $(0, 3)$. Diagonal OB is shown as a dashed line. A 20 kN vertical force acts upwards at A . A 20 kN horizontal force acts leftwards along OA towards O . A 10 kN vertical force acts downwards at C . A 20 kN force acts at B , oriented perpendicular to diagonal OB . A counter-clockwise arc represents a $40 \text{ kN} \cdot \text{m}$ moment at the center.

Detailed Solution

Step 1: Geometric Analysis of Diagonal OB

The angle θ_1 made by diagonal OB with the horizontal line OA is:

$$\tan \theta_1 = \frac{\text{Height}}{\text{Width}} = \frac{3}{4}$$

$$\theta_1 = \tan^{-1} \left(\frac{3}{4} \right) = 36.869^\circ$$

Since the 20 kN force applied at point B is perpendicular to diagonal OB , the angle θ_2 that this force makes with the horizontal line is:

$$\theta_2 = 90^\circ - \theta_1 = 90^\circ - 36.869^\circ = 53.13^\circ$$

Resolving the 20 kN force at point B :

- Horizontal component (rightward): $20 \cos(53.13^\circ) \approx 12 \text{ kN}$
- Vertical component (downward): $20 \sin(53.13^\circ) \approx 16 \text{ kN}$

Step 2: Summation of Forces

Summing horizontal forces along the x -axis:

$$\Sigma F_x = -20 + 20 \cos(53.13^\circ)$$

$$\Sigma F_x = -20 + 12 = -8 \text{ kN}$$

Therefore, $\Sigma F_x = 8 \text{ kN}$ ($\leftarrow$)

Summing vertical forces along the y -axis:

$$\Sigma F_y = 20 - 10 - 20 \sin(53.13^\circ)$$

$$\Sigma F_y = 20 - 10 - 16 = -6 \text{ kN}$$

Therefore, $\Sigma F_y = 6 \text{ kN}$ ($\downarrow$)

Step 3: Resultant Magnitude and Direction

Magnitude:

$$R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2} = \sqrt{(-8)^2 + (-6)^2} = \sqrt{64 + 36} = \sqrt{100} = 10 \text{ kN}$$

Direction relative to the horizontal:

$$\theta = \tan^{-1} \left(\left| \frac{\Sigma F_y}{\Sigma F_x} \right| \right) = \tan^{-1} \left(\frac{6}{8} \right) = 36.869^\circ$$

The resultant R acts at an angle of 36.869° pointing into the third quadrant (downward and leftward).

Step 4: Net Moment About Point O

Taking counter-clockwise moments as positive (+):

- 20 kN force at A (upward): Distance = 4 m, Moment = $+(20 \times 4) = +80 \text{ kN} \cdot \text{m}$ (CCW)
- 10 kN force at C (downward): Passes through vertical line OC ($d = 0 \text{ m}$), Moment = 0
- 20 kN force at O (leftward): Line of action passes directly through point O , Moment = 0
- Components of 20 kN force at B :
 - Horizontal component $20 \cos(53.13^\circ)$ (rightward) at height $y = 3 \text{ m}$: Moment = $-(20 \cos(53.13^\circ) \times 3) = -(12 \times 3) = -36 \text{ kN} \cdot \text{m}$ (CW)
 - Vertical component $20 \sin(53.13^\circ)$ (downward) at distance $x = 4 \text{ m}$: Moment = $-(20 \sin(53.13^\circ) \times 4) = -(16 \times 4) = -64 \text{ kN} \cdot \text{m}$ (CW)
- Pure couple moment applied at center: $+40 \text{ kN} \cdot \text{m}$ (CCW)

Summing all moments about point O :

$$\Sigma M_O^F = +80 - 36 - 64 + 40 = 20 \text{ kN} \cdot \text{m} \quad (\text{Counter-Clockwise})$$

Step 5: Perpendicular Distance from Point O

Using Varignon's Theorem:

$$\Sigma M_O^F = R \times d$$

$$20 = 10 \times d \implies d = 2 \text{ m}$$

The line of action of the resultant force $R = 10 \text{ kN}$ is located at a perpendicular distance $d = 2 \text{ m}$ from origin O .

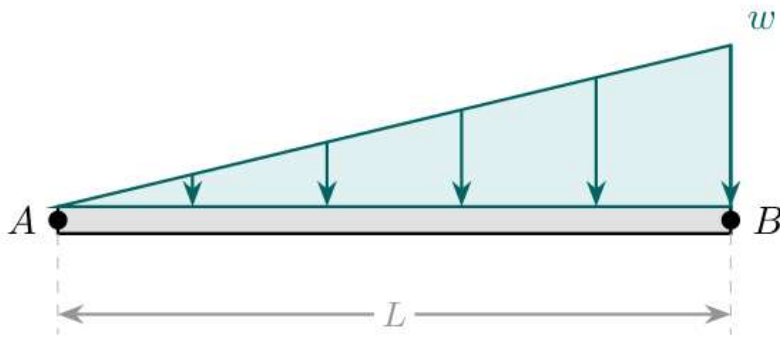
Problem 2: Vertical Frame/Pole Subjected to Inclined and Point Forces

Problem Statement

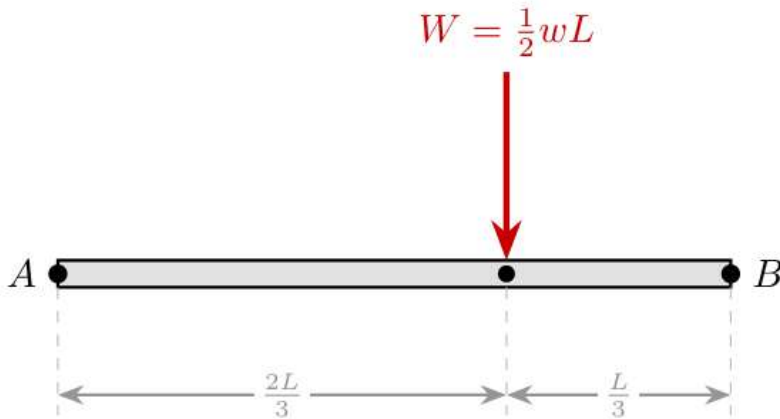
Three forces act on a vertical pole of total height 6 m fixed at base A :

1. A horizontal force of 1 kN acting leftward at height $y = 3 \text{ m}$.
2. An inclined force of 3 kN acting at the top of the pole ($y = 6 \text{ m}$) at an angle of 30° below the horizontal pointing downward-leftward.
3. A downward vertical force of 2.5 kN acting at the tip of a horizontal bracket extending 1.5 m to the right, attached at height $y = 5 \text{ m}$.

Find the magnitude, direction, and line of action of the resultant force with respect to base A .



⇓ Equivalent Conversion



Visual Description: A vertical pole attached to a fixed base at A . At top height 6 m , an inclined 3 kN force acts downwards and leftwards at 30° below horizontal. A horizontal cantilever bracket extends 1.5 m rightwards at height 5 m , carrying a 2.5 kN vertical downward force at its end. A horizontal leftward force of 1 kN acts directly on the main vertical pole at height 3 m .

Detailed Solution

Step 1: Resolving Inclined Forces

The 3 kN force at the top ($y = 6 \text{ m}$) is resolved into:

- Horizontal component (leftward): $3 \cos(30^\circ) \approx 2.598 \text{ kN}$
- Vertical component (downward): $3 \sin(30^\circ) = 1.5 \text{ kN}$

Step 2: Summation of Forces

Sum of horizontal forces (ΣF_x):

$$\Sigma F_x = -1 - 3 \cos(30^\circ) = -1 - 2.598 = -3.598 \text{ kN}$$

Therefore, $\Sigma F_x = 3.598 \text{ kN}$ ($\leftarrow$)

Sum of vertical forces (ΣF_y):

$$\Sigma F_y = -2.5 - 3 \sin(30^\circ) = -2.5 - 1.5 = -4.000 \text{ kN}$$

Therefore, $\Sigma F_y = 4 \text{ kN}$ ($\downarrow$)

Step 3: Resultant Magnitude and Direction

Magnitude:

$$R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2} = \sqrt{(-3.598)^2 + (-4)^2} = \sqrt{12.946 + 16} = \sqrt{28.946} \approx 5.38 \text{ kN}$$

Orientation angle θ :

$$\theta = \tan^{-1} \left(\frac{|\Sigma F_y|}{|\Sigma F_x|} \right) = \tan^{-1} \left(\frac{4}{3.598} \right) \approx 48.03^\circ \quad (\text{below horizontal})$$

Step 4: Moment Summation About Base Point A

1. 1 kN horizontal force at $y = 3 \text{ m}$ (leftward):

$$\text{Moment} = +(1 \times 3) = +3 \text{ kN} \cdot \text{m} \quad (\text{CCW})$$

2. 2.5 kN vertical force at horizontal distance $x = 1.5 \text{ m}$ (downward):

$$\text{Moment} = -(2.5 \times 1.5) = -3.75 \text{ kN} \cdot \text{m} \quad (\text{CW})$$

3. 3 kN inclined force components at $y = 6 \text{ m}$:

- Horizontal component $3 \cos(30^\circ)$ (leftward) at height $y = 6 \text{ m}$:

$$\text{Moment} = +(3 \cos(30^\circ) \times 6) = +(2.598 \times 6) = +15.588 \text{ kN} \cdot \text{m} \quad (\text{CCW})$$

- Vertical component $3 \sin(30^\circ)$ acts along the centerline of the pole ($x = 0 \text{ m}$):

$$\text{Moment} = 0$$

Net Moment about A:

$$\Sigma M_A^F = +3 - 3.75 + 15.588 = +14.838 \text{ kN} \cdot \text{m} \quad (\text{Counter-Clockwise})$$

Step 5: Perpendicular Distance from A

$$d = \frac{|\Sigma M_A^F|}{R} = \frac{14.838}{5.38} \approx 2.76 \text{ m}$$

Summary Equations Reference Table

Parameter	Formula	Description
Horizontal Force Sum	$\Sigma F_x = \sum F_i \cos \theta_i$	Vector sum of all x -components
Vertical Force Sum	$\Sigma F_y = \sum F_i \sin \theta_i$	Vector sum of all y -components
Resultant Magnitude	$R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2}$	Magnitude of net combined force
Resultant Angle	$\theta = \tan^{-1} \left \frac{\Sigma F_y}{\Sigma F_x} \right $	Angle relative to horizontal axis
Varignon's Principle	$\Sigma M_O = R \times d$	Equates net component moments to total resultant moment
Line of Action Offset	$d = \frac{ \Sigma M_O }{R}$	Perpendicular distance from reference point